

Referee Report:

“An Exact Algorithmic Solution of the Belgian Chocolate Problem”

Independent adversarial mathematical review

9 September 2026

1 Executive verdict and exact scope of the decision

FAIL for the frozen package of claims in the review request. The same-parameter equivalence required in items (E) and VIII is false at the endpoint. With the unchanged definitions, one has

$$\mathcal{F}_{\sqrt{Q}} \neq \emptyset, \quad \forall N \ H_N(Q) \text{ is feasible}, \quad \frac{1-Q}{1+Q} \notin \mathcal{A}.$$

Thus original polynomial admissibility is *not* equivalent to the unqualified analytic feasibility predicate used by the hierarchy.

An essential qualification. This failure is not a counterexample to the displayed Main Theorem of the manuscript. The manuscript’s own precise argument distinguishes strict analytic feasibility from feasibility at the analytic endpoint. Its Main Theorem, the hierarchy, the two strict-side certificates, the precision algorithm, and polynomial endpoint exclusion survive the mathematical checks below. The additional unqualified pointwise equivalence is imposed explicitly in the present frozen review specification, whereas the manuscript uses a strict-side converse in its Section 10.5.

The distinction is substantive and must not be concealed in either direction. It would be incorrect to report that the compactness argument fails, that the approximation algorithm has been disproved, or that the displayed interval theorem has been refuted. It would also be incorrect to certify every frozen claim as stated. Under the stipulated rule that a false frozen claim entails FAIL, that is the overall decision. If the object of review were restricted to the displayed manuscript theorem and its actual strict-interface proof, the endpoint objection would not refute that theorem.

This report does not alter the manuscript, change either endpoint convention, or supply a new research route. Section 12 gives the minimal failure certificate; Section 13 separately identifies the statement that would have to change. No modified statement is counted as passing the frozen package.

2 Audit material, freezing, and claimed theorem

The manuscript, denoted M, is the UTF-8 file

`belgian_chocolate_complete_algorithmic_solution.md`.

Its SHA-256 fingerprint at the start of this review was

B846BD95A1DB94C471BC313C319E8E535
F29BE8BC23C1B7F2F41BA3F02B296D0.

Line references below refer to this frozen file, not to this report. The original formal problem was read independently from the supplied problem-description file. Prior agents' approval statements, including M's final audit paragraph, were not used as evidence. No mathematical assertion in M was edited.

2.1 The displayed Main Theorem actually stated in M

M, lines 11–23 assert that a specified rational-arithmetic and real-algebraic algorithm produces nested rational intervals $[L_j, R_j]$, each computed in finite time, with

$$R_j - L_j \leq \frac{1}{2}(2/3)^j.$$

If Q is their unique common point and $\Delta = (1 - Q)/(1 + Q)$, then

$$\forall \delta > 0 : \quad \delta \in \mathcal{A} \iff \delta < \Delta, \quad \Delta \notin \mathcal{A}. \quad (1)$$

The theorem does not assert that Q is algebraic, that an extremal function is explicitly computed, or that equality of arbitrary real-number inputs with Q can be decided in finite time. M, lines 727–735, explicitly disclaims the last assertion. These exclusions are part of the frozen scope and are respected here.

2.2 The stronger pointwise bridge in the review specification

The present review request additionally requires the chain

$$\text{original polynomial feasibility} \iff \mathcal{F}_{\sqrt{q}} \neq \emptyset \iff \forall N \ H_N(q) \text{ feasible} \quad (2)$$

at the corresponding parameter $q = (1 - \delta)/(1 + \delta)$, without changing definitions or strictness. This is not the same assertion as (1). Distinguishing them is necessary to give an accurate referee decision rather than to manufacture a counterexample to a theorem that M does not state.

3 Reconstruction of the original problem and its analytic image

Let

$$a_\delta(s) = s^2 - 2\delta s + 1, \quad b(s) = s^2 - 1.$$

A nonzero real polynomial is Hurwitz stable when every complex zero has negative real part; nonzero constants are stable. The authoritative original predicate is

$$\delta \in \mathcal{A} \iff \exists x, y, p \in \mathbb{R}[s] \setminus \{0\} : \begin{cases} x, y, p \text{ Hurwitz stable,} \\ \deg x \geq \deg y, \\ p = a_\delta x + by. \end{cases} \quad (3)$$

There is no monic normalization, no sign restriction, and no condition $\deg p = \deg x + 2$. In particular, leading-coefficient cancellation is allowed.

The controller notation is $c = y/x$, with plant b/a_δ , and the complementary closed-loop expression is $by/(a_\delta x + by)$. Properness of c corresponds to the displayed degree condition. One must not append a separate well-posedness condition at infinity from a different control convention: the given algebraic definition is authoritative. The polynomial formulation agrees with the formulation printed in Charles–Boston, Section 1 [2]; that source also distinguishes the unrestricted problem from its fixed-degree optimization versions.

3.1 Exclusion of $\delta \geq 1$

For $\delta > 1$, a_δ has positive roots $r_- < 1 < r_+$. Equation (3) gives

$$\frac{p(r_-)}{y(r_-)} = r_-^2 - 1 < 0, \quad \frac{p(r_+)}{y(r_+)} = r_+^2 - 1 > 0.$$

The ratio of two stable real polynomials has constant nonzero sign on the positive real axis, a contradiction. At $\delta = 1$, $p(1) = 0$. This verifies M, lines 473–490, without a degree assumption on p .

3.2 Cayley transformation and parameter direction

For $0 < \delta < 1$, set

$$w = \frac{s-1}{s+1}, \quad s = \frac{1+w}{1-w}, \quad q = \frac{1-\delta}{1+\delta}, \quad t = \sqrt{q}. \quad (4)$$

The right half-plane corresponds to $|w| < 1$, and q decreases strictly with δ . Direct calculation gives

$$(s+1)^2 w = b(s), \quad (s+1)^2 (w^2 + q) = (1+q)a_\delta(s). \quad (5)$$

From a polynomial witness define

$$f(w) = \frac{b(s)y(s)}{p(s)}, \quad 1 - f(w) = \frac{a_\delta(s)x(s)}{p(s)}.$$

These are holomorphic on $\mathbb{D}$ and real symmetric. The unique zero of f there is the simple zero $w = 0$. The only two 1-points are the simple points $w = \pm it$. Stability prevents cancellations at these three points. Possible behavior at $w = 1$, which represents $s = \infty$, is irrelevant to holomorphy on the *open* disk.

Thus polynomial admissibility implies membership of $\mathcal{F}_t$, where $\mathcal{F}_t$ has exactly M’s definition: real-symmetric holomorphic functions on $\mathbb{D}$ with these three prescribed simple fibers. This implication alone is not a pointwise equivalence.

Write $\mathcal{T} = \{q \in (0, 1) : \mathcal{F}_{\sqrt{q}} \neq \emptyset\}$ and $q_* = \inf \mathcal{T}$, as in M. Nonemptiness, positivity of the infimum, attainment, and its equality with the algorithm-defined Q are audited below rather than assumed here.

3.3 Three distinct predicates

For clarity of review, without redefining M’s objects, write

$$\begin{aligned} P(q) : (1-q)/(1+q) &\in \mathcal{A}, \\ A(q) : \mathcal{F}_{\sqrt{q}} &\neq \emptyset, \\ H(q) : \forall N \ H_N(q) &\text{ feasible.} \end{aligned}$$

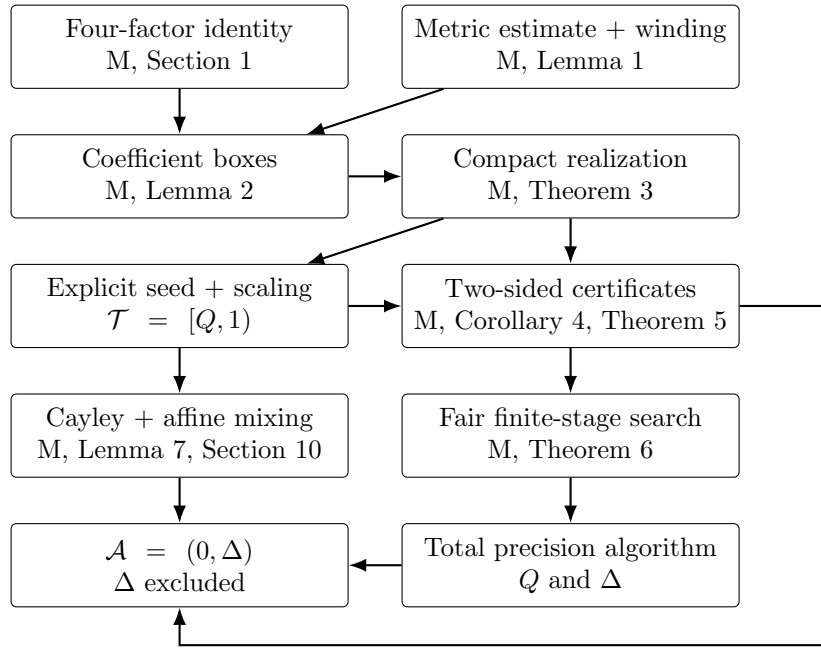
Within $0 < q \leq 9/16$, the audited conclusions are

$$P(q) \iff q > Q, \quad A(q) \iff H(q) \iff q \geq Q. \quad (6)$$

This one-point difference is the decisive issue in (2); it is not a failure of the original parameter-set conclusion (1).

4 Dependency graph of the proof

The following graph records actual dependencies rather than the order of exposition. No arrow from the polynomial endpoint theorem is needed to establish analytic endpoint attainment.



M, Section 11, is a separate consequence of coefficient-box slack and analytic endpoint attainment. It proves non-stabilization of the hierarchy and is not used to prove termination. The false shortcut $P \iff A$ is not needed by the actual displayed Main Theorem.

5 Critical-implication audit: representation and effective bounds

5.1 Four-factor representation: M, lines 46–61

The quotients

$$u = f/z, \quad v = (1 - f)/(z^2 + q), \quad U = 1/u, \quad V = 1/v$$

are holomorphic on $\mathbb{D}$: prescribed multiplicities make the apparent singularities removable and make u, v nonvanishing there. Conversely

$$zu + (z^2 + q)v = 1, \quad uU = vV = 1 \quad (7)$$

forces u and v to be nonzero everywhere. Therefore $f = zu$ has precisely the stated simple fibers. Real Taylor coefficients are equivalent to the required real symmetry. No condition on f' away from the marked fibers has entered. **Verified.**

5.2 Metric estimate and effective constants: M, lines 70–111

The normalization $\rho_{\mathbb{D}} = 2/(1 - |z|^2)$ has curvature -1 . The Hempel bound used by M is weaker than

$$\rho_{\mathbb{C} \setminus \{0,1\}}(w) \geq \frac{1}{|w|(C + |\log |w||)}, \quad C = \frac{\Gamma(1/4)^4}{4\pi^2}.$$

This normalization and specialization are independently supported by Kraus–Roth–Sugawa, Theorem 2.2, Remark 2.3 and Corollary 2.5 [4]. Setting all three singularity orders to one gives the cusp limit; their value at -1 is $4\pi^2/\Gamma(1/4)^4$. The extra factor two in M’s denominator is harmless.

The elementary estimate $\Gamma(1/4) < 4 + e^{-1} < 5$ follows by splitting its defining integral at one. Hence $C < 625/36 < 64$. The actual algorithm only uses 64 and rational data; Gamma and logarithms are not hidden coefficients in H_N . **Verified.**

5.3 Winding anchor: M, lines 137–152

The image of $|z| = R > t$ has winding numbers $(1, 2)$ about $(0, 1)$ by the argument principle. If it avoided

$$K = \{w : 1/4 \leq |w| \leq 2, |w - 1| \geq 1/4\},$$

its connected image would lie in one of three disjoint open regions: $|w| < 1/4$, $|w - 1| < 1/4$, or $|w| > 2$. Their possible winding pairs are respectively $(n, 0)$, $(0, n)$ and (n, n) . None equals $(1, 2)$. Thus a bounded, nondegenerate anchor exists on the image loop. The point need not be effectively found: its existence establishes a fixed explicit bound, and the algorithm never queries its position. **Verified.**

5.4 Propagation and maximum modulus: M, lines 115–194

For rational $r < 1$, M fixes $\beta = 3/4$ and

$$R = \frac{1 + \max(r, \beta)}{2}, \quad d = \frac{1}{2} \min(R - \beta, 1 - R), \quad J = \left\lceil \frac{16R}{d} \right\rceil + 1, \quad B = 3^{66.9^J}.$$

The radius- d disks centered on $|z| = R$ avoid $0, \pm it$ and lie in $\mathbb{D}$ for every $0 < t \leq \beta$. On each disk, both f and $1 - f$ omit $0, 1$. Schwarz–Pick and the density estimate give the stated differential inequality. Along a radial segment of length at most $d/2$, its integrated multiplier is

$$\exp\left(2 \int_0^{d/2} \frac{2d}{d^2 - s^2} ds\right) = 9.$$

The circle can be traversed in at most J such steps. Applying maximum modulus to the *four factors*, not to the meromorphic quotients $1/f$ and $1/(1 - f)$ across their poles, gives precisely the four bounds in M, lines 187–190. Their sum, rounded up, is M’s $M(r)$.

The resulting bound is uniform in the object, its degree if any, and $t \in (0, \beta]$. Its size is enormous but finite and computable. There is no use of a non-effective normal-family constant. **Verified.**

5.5 Coefficient growth: M, lines 198–224

With $r_m = 1 - 2^{-m}$, $M_m = \max(1, M(r_m))$ and

$$C_j = \min_{1 \leq m \leq j+1} M_m r_m^{-j},$$

Cauchy's estimate bounds each of the four actual coefficients by C_j . For each fixed m , the inequality $C_j \leq M_m r_m^{-j}$ holds for all $j \geq m - 1$. Consequently

$$\limsup_{j \rightarrow \infty} C_j^{1/j} \leq r_m^{-1} \quad \text{for every } m,$$

and then the upper limit is at most one. All constants are rational and computed without Q . This is a genuine tail-growth control, not a bound valid only for a fixed truncation. **Verified.**

6 Finite algebra and compact realization

6.1 The frozen system H_N : M, lines 228–242

There are exactly $4(N + 1)$ real variables u_j, v_j, U_j, V_j , $0 \leq j \leq N$. Besides the closed boxes $|h_j| \leq C_j$, the equations are

$$u_{k-1} + v_{k-2} + qv_k = \mathbf{1}_{k=0}, \quad \sum_{i=0}^k u_i U_{k-i} = \mathbf{1}_{k=0}, \quad \sum_{i=0}^k v_i V_{k-i} = \mathbf{1}_{k=0},$$

for $0 \leq k \leq N$, with negative-index coefficients zero. For rational q all data are rational. These are polynomial equations of degree at most two in the coefficient variables and linear box inequalities. No holomorphic extension, convergence, or infinite zero-free condition is left inside the finite system. Effective quantifier elimination for real closed fields decides it. **Verified.**

6.2 Inverse-limit issue: M, lines 244–266

Use the single compact product

$$\mathcal{K} = \prod_{j \geq 0} [-C_j, C_j]^4.$$

The cylinder $E_N(q)$ imposing the first $N + 1$ orders of all equations is closed. The equations at level $N + 1$ include those at level N , so $E_{N+1}(q) \subset E_N(q)$. A finite solution extends to some point of its cylinder by setting unused coordinates to zero. This does *not* assert that the zero extension solves higher-order equations.

Every $E_N(q)$ is therefore nonempty under the hypothesis. Nested compactness supplies one point in their intersection. Surjectivity of projections of finite solution sets is unnecessary. Growth of C_j gives convergence of all four power series on every compact subdisk. Absolutely convergent Cauchy products recover (7). These identities prohibit loss of zero-freeness, including degeneration to the zero function. Since $q > 0$, the two 1-points remain distinct and simple. **Verified.**

6.3 Analytic interval and endpoint attainment: M, lines 268–320

The displayed seed at $q = 1/2$ satisfies the identity exactly. Its u -roots are $-5/4, -20/17$; the roots of the quadratic factor of v are nonreal conjugates of modulus $\sqrt{1089/272} > 1$. Thus both factors extend zero-free to a slightly larger disk. Dilation yields a feasible parameter below $1/2$.

The order-zero equation gives $q \geq 1/C_0$ for every feasible $q \leq 9/16$. Hence $q_* := \inf \mathcal{T}$ is positive. If $q_0 < q < 1$ and $f_0 \in \mathcal{F}_{\sqrt{q_0}}$, then $f_0(\sqrt{q_0/q}z)$ belongs to $\mathcal{F}_{\sqrt{q}}$. This proves upper-interval feasibility without assuming an extremizer.

For feasible $q_n \rightarrow q_*$, eventually $q_n < 1/2$ and their coefficient vectors belong to the common compact product. A subsequence converges together with the parameter. Each finite equation is continuous in these coordinates and q . The realization proof applies at the positive limit. Thus

$$0 < q_* < 1/2, \quad \mathcal{T} = [q_*, 1), \quad \mathcal{F}_{\sqrt{q_*}} \neq \emptyset. \quad (8)$$

No polynomial endpoint theorem is used in this derivation. **Verified.**

6.4 Finite negative certificates: M, lines 322–329

Within the stated range, Theorem 3 and (8) imply

$$q < q_* \iff \neg(\forall N H_N(q)) \iff \exists N \neg H_N(q).$$

The system being negated is exactly the frozen system. This is ordinary classical quantifier negation after a valid realization theorem, not an additional compactness assumption. A negative certificate excludes all degrees, not merely degree N . At $q = 0$, $qv_0 = 1$ is already inconsistent. **Verified.**

7 Strict positive certificates and rational approximation

M, lines 334–377. For rational $0 < q \leq 1/2$, the certificate is a rational polynomial v with $v(0) = 1/q$ such that

$$u(z) = \frac{1 - (z^2 + q)v(z)}{z}$$

and u, v are zero-free on $\overline{\mathbb{D}}$.

Exact interpolation. The fixed constant term makes the quotient a polynomial. Defining u from v , instead of approximating them independently, preserves the identity exactly.

Strict margin. If $q > q_*$, choose a feasible $q_0 < q$. Dilation produces analytic factors on a disk of radius $\sqrt{q/q_0} > 1$. They have positive minimum moduli on $\overline{\mathbb{D}}$. Approximate v there by real Taylor polynomials and rationally approximate the finitely many nonconstant coefficients, preserving its rational constant term.

For the induced u_n , on $|z| = 1$,

$$|u_n - u| \leq (1 + q)\|v_n - v\|_{\overline{\mathbb{D}}}.$$

The error is holomorphic at zero, so maximum modulus transfers the bound to the whole disk. Taking the approximation sufficiently close preserves both positive minimum moduli. A non-effective choice of a successful index is acceptable in this *existence* proof because the algorithm independently enumerates and verifies every finite candidate.

Verification. For a rational polynomial h , writing $h(x + iy) = A(x, y) + iB(x, y)$ reduces closed-disk zero-freeness to

$$\neg \exists x, y \in \mathbb{R} : x^2 + y^2 \leq 1 \wedge A(x, y) = B(x, y) = 0.$$

This is finite real algebra. It rejects the identically zero polynomial automatically.

Converse strictness. Any such certificate has a zero-free extension to some disk of radius $R > 1$; dilating zu gives feasibility at $q/R^2 < q$. Hence $q > q_*$. A positive certificate cannot exist at $q = q_*$. All steps preserve real symmetry and rational coefficients when q is rational. **Verified.**

8 Algorithm termination audit

M, lines 387–454. The domain invariant is $0 \leq L < q_* < R \leq 1/2$. Put

$$a = (2L + R)/3, \quad b = (L + 2R)/3.$$

Then $0 < a < b < 1/2$. Each stage s first decides the two finite systems $H_s(a), H_s(b)$, and then checks two finite lists of positive-certificate candidates of bounded degree and rational height. Every individual test terminates. Repeating candidates is inefficient but harmless.

Audit obligation	Finding
Distinct test points	$b - a = (R - L)/3 > 0$.
No equality oracle	At most one point equals q_* . No branch tests this equality.
Exhaustive strict sides	Every other test point is either below q_* , with a finite negative layer, or above it, with a finite positive witness.
Fairness	A candidate of degree d whose reduced coefficients have numerator magnitudes and positive denominators at most h appears by stage $s \geq \max(d, h)$. Before any fixed stage, only finitely many terminating tests occur.
Safe update	Negative evidence returns $[a, R]$ or $[b, R]$; positive evidence returns $[L, a]$ or $[L, b]$. In all cases the new bounds remain strict.
Effective contraction	The four width ratios are $2/3, 1/3, 1/3, 2/3$.
Prescribed precision	Choose j by rational arithmetic so $\frac{1}{2}(2/3)^j < 2^{-m}$. The finite composition of j terminating refinements produces the requested strict enclosure.

The implementation is a fair finite-stage dovetailing, not a sequential wait for an infinite unsuccessful search. Endpoint ambiguity does not make both test points stall. The strict $< 2^{-m}$ requested in this review follows by choosing j with strict width inequality; no frozen algorithm needs changing. **Verified.**

The unique intersection point is then the same q_* , so the independently algorithm-defined Q equals it. The transformed δ -interval has width

$$\frac{2(R - L)}{(1 + L)(1 + R)} \leq 2(R - L).$$

No numerical experiment or finite output is being identified with the exact endpoint.

9 Endpoint audit, cancellation, and the reverse polynomial construction

9.1 Fixed-degree stability and affine mixing

M, lines 465–553. Stability is open at fixed degree, by disjoint small circles around the roots and Rouché’s theorem. If $\deg y < \deg x = n$, then p has degree $n + 2$ with unchanged nonzero leading coefficient under a small change of δ . The direct perturbation argument is valid in this case.

When $\deg y = \deg x = n$, M instead sets

$$k = \sqrt{1 + \varepsilon^2 + 2\varepsilon/\delta}, \quad \delta' = \frac{k\delta}{1 + \delta\varepsilon}, \quad \phi(s) = ks - \varepsilon,$$

$$\begin{aligned} A &= 1 + \varepsilon^2 + \varepsilon(\delta + \delta^{-1}), & B &= \varepsilon^2 + \varepsilon/\delta, \\ C &= \varepsilon(\delta^{-1} - \delta), & D &= 1 + \varepsilon/\delta. \end{aligned}$$

The identities

$$a_\delta \circ \phi = Aa_{\delta'} + Cb, \quad b \circ \phi = Ba_{\delta'} + Db$$

were checked independently by coefficient comparison and exact symbolic algebra. Thus

$$X = A(x \circ \phi) + B(y \circ \phi), \quad Y = C(x \circ \phi) + D(y \circ \phi), \quad P = p \circ \phi$$

satisfy $P = a_{\delta'}X + bY$. The polynomials X, Y have degree at most n and their leading coefficients tend to the original nonzero ones. The degree of P is exactly the actual degree of p . Therefore small $\varepsilon > 0$ preserves all three degrees and all three stability conditions. Also

$$1 - \delta'^2 = \frac{1 - \delta^2}{(1 + \delta\varepsilon)^2}$$

proves $\delta < \delta' < 1$. The lemma does not rely on a root at infinity remaining stable; it avoids changing the number of finite roots. **Verified.**

9.2 A genuine cancellation stress test

Take

$$\delta = \frac{1}{2}, \quad x = s + \frac{3}{2}, \quad y = -s - \frac{1}{2}, \quad p = \frac{1}{2}s + 2.$$

All three are strictly stable, $\deg x = \deg y = 1$, and $p = a_{1/2}x + by$. The nominal cubic expression has dropped to degree one. Keeping x, y fixed and changing δ to $1/2 + \eta$ gives

$$p_\eta = -2\eta s^2 + (1/2 - 3\eta)s + 2.$$

For every $\eta > 0$ the product of its roots is negative, so one root is positive. This destroys the naive openness argument.

It does *not* destroy M's actual lemma. At $\varepsilon = 1/100$, its transformation has $k = \sqrt{10401}/100 > 0$ and yields exactly

$$X = \frac{201k}{200}s + \frac{30351}{20000}, \quad Y = -\frac{201k}{200}s - \frac{9549}{20000}, \quad P = \frac{k}{2}s + \frac{399}{200}.$$

Each is stable and degree one. This is a relevant adversarial test, rather than a generic full-degree example.

9.3 Strict necessity and exclusion of the polynomial endpoint

M, lines 555–589. Cayley transformation of a witness at δ gives $q(\delta) \geq q_*$. The perturbation lemma gives another witness at $\delta' > \delta$, so

$$q(\delta) > q(\delta') \geq q_*.$$

Thus $\delta \in \mathcal{A}$ implies $q(\delta) > q_*$. Endpoint exclusion is a theorem, not a convention about suprema. In particular the analytic extremizer from (8) cannot correspond to an original polynomial witness at the same endpoint.

9.4 Strict sufficiency and degree preservation

M, lines 591–640. For real $q \in (q_*, 1)$, the strict positive-factor argument works with real Taylor truncation. Let u, v be the resulting closed-disk zero-free polynomials, $n = \max(\deg u, \deg v)$, and use (4). Set

$$x = (1 + q)(s + 1)^n v(w), \quad y = (s + 1)^n u(w), \quad p = (s + 1)^{n+2}.$$

Equation (5) proves the original identity. Since $u(1)v(1) \neq 0$, both x and y have degree exactly n , not just at most n . Every root ω of u or v has $|\omega| > 1$ and maps to

$$\operatorname{Re} \frac{1 + \omega}{1 - \omega} = \frac{1 - |\omega|^2}{|1 - \omega|^2} < 0.$$

Any roots added in clearing denominators are -1 . The construction gives all three nonzero stable polynomials with the required degree order. This establishes the reverse implication for $q > q_*$, exactly as M states it; it does not establish it for $q = q_*$. **Verified.**

10 Non-finite stabilization and adversarial countermodels

10.1 The actual hierarchy does not stabilize

M, lines 642–703. The parameter sets S_N are nonempty compact semialgebraic projections of compact sets. Their minima α_N are algebraic and can be found by real quantifier elimination and root isolation. Nestedness and the realization theorem give $\alpha_N \uparrow q_*$. Closedness of each fixed S_N justifies passage to this limit.

Each endpoint coefficient lies strictly inside its box: $M(r)$ is the sum of four positive individual bounds, so it strictly exceeds each individual bound, and the minimum defining C_j is finite. For nearby q , the germs

$$v_q = v_* + 1/q - 1/q_*, \quad u_q = \frac{1 - (z^2 + q)v_q}{z}$$

have $v_q(0) = 1/q$ and $u_q(0) = (q/q_*)u_*(0) \neq 0$. Their inverse germs have finite jets continuous in q . Hence every fixed collection of equations and boxes remains feasible for q on both sides of q_* . Taking the neighborhood small enough to stay in $(0, 1/2)$ gives $\alpha_N < q_*$ for every finite N .

The germs need not extend as units to $\mathbb{D}$ when $q < q_*$. The proof does not make that claim. It correctly disproves permanent stabilization of this particular hierarchy, without disproving other possible finite characterizations. **Verified.**

10.2 Attempted counterexamples and their disposition

Attack	Disposition
Each finite level is feasible but no infinite object exists	The abstract closed sets $[N, \infty) \subset \mathbb{R}$ exhibit this without compactness. M instead works in one fixed compact product. The countermodel does not satisfy its hypotheses.
Compact ambient space but no closed constraints	The nested sets $(0, 1/N)$ in $[0, 1]$ have empty intersection. M uses polynomial equalities and closed boxes, so this attack is inapplicable.
Zero-free functions collapse to zero	$h_n \equiv 1/n$ does this, but $1/h_n = n$ violates the uniform inverse bound. M controls both factors and inverses.
An interior zero emerges	The limit identity $uU = 1$ forbids it at every point of $\mathbb{D}$. Boundary zeros are not excluded for a general analytic limit and are not needed by realization.
A boundary zero prevents rational approximation	Strict-side dilation provides a larger zero-free disk before approximation. At the endpoint no such strict certificate is asserted.
Rational rounding destroys interpolation	Only v is approximated, its constant is fixed, and u is defined algebraically. The exact identity is retained.
Transcendental data occur in CAD	Gamma enters only a proved estimate and is replaced by 64. The finite systems and rational positive-certificate tests use rational constants.
Both third-points stall at equality	Two distinct numbers cannot both equal Q . Fair stages guarantee the other point is reached and certified.
Degree cancellation defeats endpoint perturbation	The explicit stress test above defeats naive perturbation but is preserved by the actual affine-mixing lemma.
The raw analytic predicate is equivalent to the original predicate	This attack succeeds at $q = Q$. It is the minimal failure of the additional frozen chain, not a compactness counterexample.

10.3 Independent finite calculations

Exact symbolic calculations were performed with Python 3.12 and SymPy 1.14.0. The checks were polynomial/rational simplifications, not floating-point root tests:

Expression checked	Result
$a_\delta(ks - \varepsilon) - Aa_{\delta'}(s) - Cb(s)$, using $k^2 = 1 + \varepsilon^2 + 2\varepsilon/\delta$	0
$b(ks - \varepsilon) - Ba_{\delta'}(s) - Db(s)$, with the same substitution	0
$1 - \delta'^2 - (1 - \delta^2)/(1 + \delta\varepsilon)^2$	0
Both identities in (5)	0
$zu + (z^2 + 1/2)v - 1$ for M's explicit seed	0
Discriminant of $1089 + 1000z + 272z^2$	-184832
Original polynomial residual for the transformed seed at $\delta = 1/3$	0

The full large-constant CAD hierarchy was not executed, and no computed digits of Q are claimed. These finite checks corroborate the displayed algebra; they do not replace the analytic proofs.

11 Literature consistency check

The literature was searched independently during this review. Primary sources, rather than earlier agent summaries, were used for the following comparisons. Publication dates are those of the papers, not web-crawler dates. The search did not locate an earlier identical two-sided coefficient-hierarchy algorithm or a verified later improvement to the numerical records listed below; this is a bounded search finding, not a proof of priority or an exhaustive bibliographic claim.

11.1 Original formulation and numerical records

Charles–Boston [2], published in 2018, prints the same polynomial identity and degree restriction and reports admissibility through $\delta = 0.9808348$. Its structural optimization does not claim an unrestricted sharp endpoint theorem. Bergweiler–Eremenko [1], Section 11, gives the upper estimate $\delta_* < 0.999579$ and describes the polynomial interval as open at its upper endpoint, while the holomorphic interval includes its lower endpoint. Thus the one-point distinction in (6) agrees with the classical source, not with the unqualified bridge (2).

These record values are literature checks, not computations made by the candidate algorithm, and no rounded decimal is treated as an exact critical value. The manuscript supplies no practical new numerical improvement.

11.2 Goldberg constants and subordination

Bergweiler–Eremenko [1], Section 1, prove a subordination result for a two-fiber configuration with multiplicities, but expressly do not extend it to one simple zero and two simple 1-points. Their conjectured extremizer for the second Goldberg constant is not a theorem identifying the Belgian Chocolate endpoint. M does not invoke that conjecture or that subordination. Its metric argument uses Schwarz–Pick for arbitrary holomorphic maps. Accordingly this known limitation does not refute its coefficient compactness argument.

11.3 Meaning of rational undecidability

Blondel–Gevers [3] use “rationally undecidable” for the absence, for the general simultaneous-stabilization problem, of a necessary-and-sufficient expression formed from finitely many arithmetic operations, logical combinations and sign tests on real plant coefficients. This is not, merely by its wording, a theorem that the threshold of this fixed one-parameter family is a noncomputable real.

M allows unbounded enumeration depth and variable-dimensional real-algebraic calculations, and its total algorithm takes a requested *precision*, not an arbitrary plant or an arbitrary exact-real equality query. Thus the cited result is not contradicted. This explanation does not reclassify rational undecidability as a Turing undecidability theorem, and it does not claim that M decides general simultaneous stabilization.

11.4 Exact real versus closed form

An explicitly specified total precision algorithm is a mathematically legitimate exact name of a real number. In conjunction with a proved identity $\mathcal{A} = (0, \Delta)$ and endpoint exclusion, it is more than a sequence of unproved bounds or a bare existential computability assertion. M’s

displayed Main Theorem can therefore be read consistently as an exact algorithmic threshold characterization under the original problem's stated no-closed-form requirement.

It does not compute an extremal function, establish algebraicity or transcendence, or establish publication novelty. Nor does it automatically give a uniform membership decision algorithm for every rational input: M does not prove Δ irrational, and the two certificate searches have no certificate at the endpoint if that endpoint is rational. For arbitrary Cauchy names of real inputs, a total membership test at the endpoint would in addition violate continuity. Neither stronger decision assertion is the displayed Main Theorem. These distinctions must remain explicit; they are not reasons to falsely disprove that theorem.

12 Minimal failure certificate for the frozen pointwise bridge

Proposition 1. *With M 's unchanged definitions, the implication*

$$\mathcal{F}_{\sqrt{q}} \neq \emptyset \implies \frac{1-q}{1+q} \in \mathcal{A}$$

is false. Consequently the original polynomial predicate is not equivalent to $\forall N H_N(q)$ on the full domain of the hierarchy.

Proof. The independent compactness verification above establishes $0 < Q < 1/2$ and $\mathcal{F}_{\sqrt{Q}} \neq \emptyset$. Theorem 3 then yields $H_N(Q)$ for every N .

Suppose nevertheless that $\Delta = (1 - Q)/(1 + Q)$ were admissible. The verified perturbation lemma gives an admissible $\delta' > \Delta$. Its Cayley image would be an analytic feasible object at

$$q' = \frac{1 - \delta'}{1 + \delta'} < Q,$$

contradicting the definition and verified attainment properties of the analytic threshold. Hence $\Delta \notin \mathcal{A}$. At $q = Q$, the antecedent of the claimed implication is true and its consequent is false. $\square$

The certificate uses no numerical approximation to Q , no conjectured structure of an extremizer, and no assumption that Q is rational. The requested original-interface chain quantifies over the real parameters of the original problem, so a real endpoint counterexample suffices. It is also precisely at a point in the domain $0 < q \leq 9/16$ of the frozen hierarchy.

The shortest form of the failure is

$$\boxed{H(Q) = A(Q) = \text{true}, \quad P(Q) = \text{false}.}$$

The manuscript itself contains the two correct endpoint assertions at lines 314–318 and 579–640. The objection does not arise from disregarding its endpoint proof; it arises from taking that proof seriously while also testing the additional pointwise equivalence demanded in the frozen review request.

13 What would have to change; not a repair credited in this audit

The raw bridge (2) cannot be repaired while preserving all its predicates and quantifiers. One must at least change its statement. The correct connections, already supported by M 's actual

strict-side argument, are

$$\begin{aligned} A(q) &\iff q \geq Q, \\ P(q) &\iff q > Q \iff \exists r \in (0, q) : \mathcal{F}_{\sqrt{r}} \neq \emptyset. \end{aligned}$$

For $0 < q \leq 9/16$, this can also be written as

$$P(q) \iff \exists r \in (0, q) \forall N : H_N(r) \text{ feasible},$$

whereas $A(q) \iff \forall N H_N(q)$ remains unchanged. This displays exactly the quantifier and strict-margin difference; it is not an invitation to new research or a replacement accepted under the frozen rules.

No change to M's displayed Main Theorem, Q , H_N , the precision algorithm, or the stability definition is needed to *state* the already-established strict-interface result accurately. What must be withdrawn is the extra unqualified pointwise equivalence in the frozen claim package. Alternatively, changing the analytic predicate to require a strict extension margin would change a frozen function class and so also could not count as an as-is pass.

14 Final referee decision

FAIL under the submitted frozen requirements. The unique decisive failure found is the endpoint counterexample to the additional same-parameter polynomial–analytic–hierarchy equivalence. It is not a nontermination counterexample, a failed inverse limit, a false coefficient bound, or a failure of the affine perturbation lemma.

The distinction between the two review objects is the final conclusion:

- The manuscript's precisely displayed *algorithmic threshold theorem* is supported by the audited proof. The absence of a closed form or a useful runtime estimate does not, by itself, invalidate that theorem or turn its proved enclosure algorithm into numerical guesswork.
- The *entire frozen package required in the present request* cannot receive the stipulated PASS, because it additionally requires a pointwise equivalence that the correct endpoint results disprove. Calling that equivalence “strict analytic feasibility” without changing its statement would conceal exactly the forbidden change of predicate.

There is therefore no basis for the unqualified certification sentence requested for PASS over all the frozen claims. Conversely, this report must not be cited as showing that equation (1) is false or that only one-sided numerical bounds were proved. Mathematical correctness of that theorem, correctness of the stronger frozen package, and novelty or journal significance are separate questions.

References

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- [3] V. Blondel and M. Gevers, *Simultaneous stabilizability of three linear systems is rationally undecidable*, Mathematics of Control, Signals, and Systems **6** (1993), 135–145. <https://doi.org/10.1007/BF01211744>. The publisher’s abstract explicitly states the intended meaning of rational undecidability.
- [4] D. Kraus, O. Roth and T. Sugawa, *Metrics with conical singularities on the sphere and sharp extensions of the theorems of Landau and Schottky*, Mathematische Zeitschrift **267** (2011), 851–868. <https://doi.org/10.1007/s00209-009-0649-x>. Public preprint, Theorem 2.2, Remark 2.3 and Corollary 2.5: <https://arxiv.org/abs/0911.0758>.